
Modern geometric graph models have overcome many previously known limitations

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Abstract

Expressivity and theoretical properties of graph networks have garnered much attention in recent years, with many strong results for graph neural networks paving the way for future architectures which address those limitations. While there has been research into expressivity of geometric GNNs, there has been no formal study into how modern geometric GNNs are affected by the known pathologies of GNNs. Given recent advancements in the Pareto frontier, it is a prime time to investigate how well modern geometric GNNs can model long-range dependencies, how badly they suffer from oversmoothing, whether they are similarly asymptotically constant, and validate their expressivity. To do this, I perform an empirical and theoretical analysis of geometric models, and replicate the results of Joshi et al.. I hope that these experiments will help direct future model architects in fruitful directions; and improve our understanding of what is causing recent empirical improvements; whether they are due to overcoming known problems, better inductive biases, or better hyperparameter selection. I undertake a theoretical analysis of geometric models, focusing on how badly geometric models suffer from the known limitations of GNNs, and to what extent equivariant transformers have overcome them.

1 Why is it important to know the limitations of geometric GNNs?

Understanding the limitations of neural architectures allows us to design better architectures which do not suffer from these limitations. Understanding the exponential forgetting problem in RNNs led to the advent of GRNNs, and Transformers [Chung et al., 2014, Vaswani et al., 2017], the latter being one of the most significant scientific inventions in recent decades.

Graph learning has particularly strong theoretical results, with the expressivity of graph neural networks (GNNs) being heavily studied in recent years [Xu et al., 2019, Oono and Suzuki, 2020, Alon and Yahav, 2021]. GNNs were discovered to suffer from many fundamental limitations, and solving these limitations has directed research in the field as a whole. Realising the limitations of message-passing neural networks led to the advent of graph transformers, and understanding the limitations of graph transformers led to the advent of structural embeddings [Kreuzer et al., 2021]. Understanding the limitations of graph transformers with structural embeddings inspired the creation of the current state-of-the-art GPS-like models [Rampasek et al., 2022]. When a new method is created, subsequent work investigates its properties, its limitations and proposes solutions. Theoretical research is, thus, essential for progress in architecture design and elsewhere.

Learning on geometric graphs is a comparatively young field, with few theoretical results, empirical results, or datasets. Despite there being much interest in this field, with many new architectures being proposed [Schütt et al., 2017, Satorras et al., 2021, Brandstetter et al., 2022, Liao and Smidt, 2023, Cen et al., 2024, Aykent and Xia, 2025], there is comparatively little research into the limitations of previous architectures. Existing theoretical research [Schütt et al., 2021, Joshi et al., 2023] only scratches the surface of these limitations, leaving room for rich investigations into oversquashing [Alon and Yahav, 2021], oversmoothing [Li et al., 2019], and other problems of GNNs which

may transfer to geometric GNNs [Adam-Day et al., 2023]. Investigating these problems could help guide future model architects, and prevent the misconception of architecture progress due to better machinery or hyperparameters [Liu et al., 2022, Tönshoff et al., 2024, Tori et al., 2025].

In this paper, I rigorously investigate the theoretical and practical limitations of modern geometric graph architectures. I investigate whether modern architectures are asymptotically constant, how well architectures can model long range dependencies, as well as investigating oversmoothing and expressivity. To allow for comparison and insight, I perform these investigations across both geometric GNNs, and equivariant transformers.

2 How can we study the limitations of geometric GNNs?

Using handcrafted tasks and datasets which are designed to elicit specific behaviours will allow me to evaluate to what extent modern geometric GNNs suffer from the limitations of GNNs. I’m particularly interested in how modern SOTA architectures compare with older architectures, informing us whether recent performance improvements are due to overcoming these limitations or not. To do this, I investigate their ability to model long-range dependencies, how models behave as the number of nodes or layers tends to infinity, and conclude this section with an investigation into expressivity, comparing expressivity of different models to the GWL test.

2.1 Investigate the effect of known pathological phenomena

GNNs are known to suffer from *many* pathological cases, such as under-reaching, oversmoothing, oversquashing, and others. These prevent GNN model scaling, limit their performance, and prevent many methods being applied on real-world data. These same limitations were assumed to hold back geometric GNNs. This assumption has, however, never been tested, and may be untrue for the most recent geometric GNNs which are very different from the early GNNs on which these pathological problems were identified. It’s therefore unclear to what extent geometric GNNs are susceptible to these limitations, and whether novel architectural decisions, such as residual connections and layer normalisation, partially mitigate these problems.

Long-range dependencies The message-passing paradigm, which geometric GNNs are based on, only allows information to flow from nodes to their immediate neighbours. For a network with depth d , nodes which are separated by more than d edges will be unable to communicate. In reality, information transfer is not perfect, meaning that a GNN may require more than d nodes to pass information between nodes which are d apart. I construct a synthetic dataset which GNNs of depth d can theoretically solve, and investigate how deep different GNN architectures have to be to actually solve this problem. This helps me answer the question ‘how much deeper than theoretically necessary do geometric GNNs need to be in order to transfer information’?

How do models behave on large graphs? To classify over graphs, GNNs first construct node features, then aggregate these node features. This pooling has recently been shown that this leads predictions on *all* graphs to tend to the same value [Adam-Day et al., 2024]. This phenomenon has never been studied in the context of Geometric GNNs. Geometric GNNs focus on equivariance and invariance, and it is not known how these properties interact with the rate of convergence to constant. I carry out a thorough investigation into whether geometric GNNs converge to constant functions on large graph, and how the rate of convergence varies between architectures to answer ‘how do geometric GNNs behave on large graphs?’

Oversmoothing GNN layers compute a form of laplacian smoothing. Repeated application of smoothing leads to ‘oversmoothing’, where all node features collapse to the same value [Li et al., 2018]. This is known to severely limit performance of deep (geometric) GNNs. Combined with under-reaching, GNNs which are too shallow, or too deep, have serious limitations. Oversmoothing is, however, known to be ameliorated by residual connections, jump connections, normalisation, and many other methods which are common in modern architectures [Rusch et al., 2023]. To answer the question ‘is oversmoothing is still a significant problem for geometric GNNs’, I empirically investigate how badly modern GNNs suffer from oversmoothing. I use random graphs and investigate how the Lipschitz Energy of the logits changes as model depth increases.

2.2 Investigate theoretical expressivity

It is well-known that any GNN based exclusively on first-order message-passing has its expressivity upper bounded by the 1-WL graph isomorphism test [Morris et al., 2019]. This means that any architectural adaptation which is still based on first-order message passing cannot possibly have higher expressivity, which has motivated research into more expressive architectures [Morris et al., 2019, Kreuzer et al., 2021].

A similar result was proved for the GWL test for equivariant geometric GNNs which *only* use message-passing. The rapid development of geometric GNNs has led to the adoption of structural embeddings [Aykent and Xia, 2025], with potential for future networks to include global readout or higher order features. This violates the assumptions in the proof that GWL upper bounds the expressivity of geometric GNNs. As such, when modern models use anything which could break out of GWL, such as GöttenNet’s use of structural encodings, it’s essential to check whether it *has* increased expressivity to guide future model architects.

3 What experiments should I run?

In this section, I explicitly state and justify my methodology. I explain the theoretical frameworks I use, and motivate the experiments which I run. To maximise the utility of my experiments for future model architects, I carry out all experiments on a wide range of geometric GNNs [Jing et al., 2021, Satorras et al., 2021, Liao and Smidt, 2023], including modern equivariant transformers, and comparatively old geometric GNNs.

3.1 Probe how well different architectures model long-range dependencies

The message-passing nature of geometric GNNs means that networks of depth k can only see nodes less than k steps away: message-passing neural networks have a fixed horizon. Inspired by the experiments of Joshi et al., I construct a set of pairs of graphs, where each pair is WL indistinguishable, and GWL distinguishable within $\lfloor \frac{k}{2} \rfloor + 1$ iterations. I train each architecture to distinguish between the pair of graphs, and record the accuracy on this task. This allows me to test how deep different geometric GNN architectures must be to distinguish these graphs. These results help quantify the empirical gap between theoretical reachability, and empirical learnability, and provide provisional figures on how good different geometric models are at modelling long-range geometric dependencies.

The chains dataset I replicate the dataset used by Joshi et al. to investigate under-reaching. This dataset consists of ‘chains’ of length k which are structurally equivalent, but geometrically distinct. The chains require at least $\lfloor \frac{k}{2} \rfloor + 1$ steps for the GWL graph isomorphism test to distinguish between. I provide an example of such graphs in fig. 1.

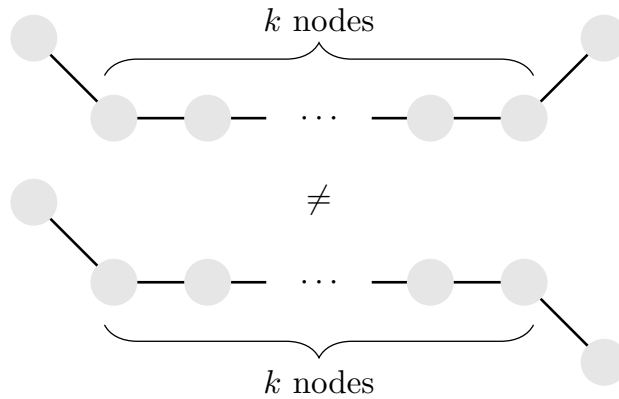


Figure 1: Visualisation of the k -chains dataset from [Joshi et al., 2023]. Geometric GNNs based on message passing cannot distinguish between these graphs unless they have at least $\lfloor \frac{k}{2} \rfloor + 1$ layers. Many geometric GNNs, however, require significantly more layers than this. This dataset highlights the difference between theoretically distinguishable graphs, and empirically distinguishable graphs.

3.2 Discover how the behaviour of networks differs as the number of nodes tends to infinity

Investigating asymptotic behaviour An understanding of the behaviour of neural networks when applied to large graphs is vitally important; as the field develops, the size of the tasks they are applied to scales as well. In light of this, I construct random Erdős Rényi graphs with random features and positions; a geometric variant of the methodology of Adam-Day et al., and investigate the distribution of predictions as graph size varies. It is known that, for conventional GNNs, models must converge to a constant function on all graphs. This result is, however, not proved for either graph transformers, geometric GNNs, or equivariant transformers. My experiments pave the way for determining whether the latter classes of models are also asymptotically constant.

3.3 Discover how susceptible modern GNNs are to oversmoothing

Each layer of a GNN computes a form of trainable Laplacian smoothing, applying too many layers leads to ‘oversmoothing’, where all latent representations are the same, and invariant of the starting node. There has been much progress on this problem recently, through residual connections and normalisation. To investigate how susceptible different architectures are to oversmoothing, I investigate the dirichlet energy [Zhou et al., 2021] of different architectures when applied to Erdős Rényi graphs with random features.

3.4 Validate model expressivity is bounded by GWL

As mentioned in section 2.2, I use the GWL test to measure the theoretical expressivity of geometric GNNs. To do this, I explicitly construct a dataset consisting graphs which are GWL indistinguishable, or IGWL indistinguishable, such as the ones in fig. 2, and test whether each geometric model is able to differentiate between them.

For a graph of size n , the GWL and IGWL graph isomorphism tests are guaranteed to converge within n iterations. By ensuring that each model has at least as many layers as the number of nodes in the graph, I ensure that results are genuine, and not artefacts of shallow models. For each architecture, I therefore construct a model deeper than the number of nodes in the graph, and test whether the logits output for each graph are different.

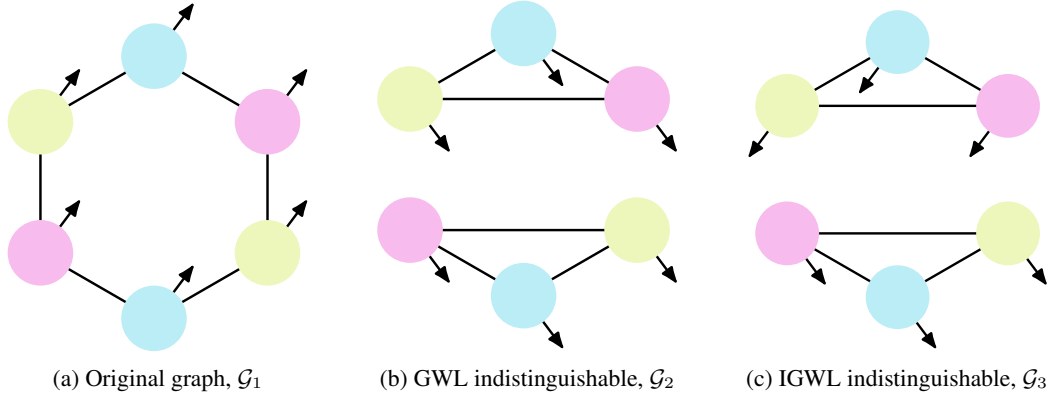


Figure 2: Example of graphs which are indistinguishable by different algorithms. Only graphs 2a and 2b are GWL indistinguishable, whilst graphs 2a, 2b and 2c are all IGWL indistinguishable. Geometric features are denoted by an arrow, whilst scalar features are denoted by colour. GWL allows for geometric features to be equivalent up to symmetry, whilst IGWL allows for geometric features within connected components to be equivalent up to symmetry.

3.5 Experimental setup

Empirical setup For all experiments in section 4, I use models with 64 hidden dimensions, and a learning rate of $1e-3$. When the experiment does not involve a specific number of layers, I use models of depth 4. I re-run oversmoothing, and asymptotically constant experiments 100 times, using 100 graphs and 10 random initialisations to ensure my results have low variability, except for

Equiformer which I only re-run 9 times due to compute requirements. I use EGNN [Satorras et al., 2021] and GVP-GNN [Jing et al., 2021] as examples of geometric GNNs, and use Equiformer [Liao and Smidt, 2023] as an equivariant transformer against which to compare.

4 Modern geometric architectures have overcome many well-known limitations of GNNs

In this section, I report the results of my experiments. I find that, whilst the expressivity of modern geometric models is still bounded by GWL, many empirical pathologies are significantly less limiting than they used to be. I conject that this is because of the intelligent design of modern architectures, *i.e.* large usage of both residual connections [He et al., 2016] and normalisation [Ioffe and Szegedy, 2015, Ba et al., 2016].

4.1 The disparity between empirical reachability and theoretical reachability is small

I replicate the experiments of Joshi et al., as described in section 3.1 and find that many architectures have a significantly lower gap than presented by Joshi et al., showing little evidence of oversquashing on this simple dataset. I present my results in table 1

Table 1: Replication of the results of [Joshi et al., 2023], updated with the performance of Equiformer [Liao and Smidt, 2023], and GotenNet [Aykent and Xia, 2025]. Notice that GotenNet is not bounded by the GWL isomorphism test, but still struggles to represent long-range dependencies.

$(k = 4\text{-chains})$		Number of layers				
GNN Layer		$\lfloor \frac{k}{2} \rfloor$	$\lfloor \frac{k}{2} \rfloor + 1 = 3$	$\lfloor \frac{k}{2} \rfloor + 2$	$\lfloor \frac{k}{2} \rfloor + 3$	$\lfloor \frac{k}{2} \rfloor + 4$
Equiv.	GWL	50%	100%	100%	100%	100%
	EGNN	50 $\pm$ 0	50 $\pm$ 0	85 $\pm$ 23	90 $\pm$ 20	100 $\pm$ 0
	GVP-GNN	50.0 $\pm$ 0.0	100 $\pm$ 0	100 $\pm$ 0	85 $\pm$ 23	100 $\pm$ 0
	GotenNet	55 $\pm$ 15	55 $\pm$ 15	50 $\pm$ 0	50 $\pm$ 0	50 $\pm$ 0
	Equiformer	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0	100 $\pm$ 0
	IGWL	50%	50%	50%	50%	50%
Inv.	SchNet	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0
	DimeNet	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0
	SphereNet	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0	50.0 $\pm$ 0.0

In table 1, my results are sometimes stronger than those of Joshi et al., *i.e.* EGNN. I believe that this is because I used Adam’s recommended learning rate of $1e-3$, rather than $1e-4$, with a scheduler decreasing it to $1e-5$, as Joshi et al. did. These results indicate that most modern geometric models still suffer from significant discrepancies between empirical reachability and theoretical reachability. Many models struggle to model long-range dependencies, even when they can theoretically express them. This highlights that modern models still struggle to represent long-range dependencies, and may be the reason why learning on heterophilic graphs remains such a challenge.

These results demonstrate that Equiformer, which is based on the transformer architecture, can distinguish graphs faster than the GWL algorithm can. This is because Equiformer allows nodes to attend to nodes which they are not directly connected to. Notice that this does not imply that Equiformer is more expressive than GWL, only that it converges faster than GWL does.

4.2 All geometric models tend to constant functions

Following the work of Adam-Day et al., I investigate whether geometric models are asymptotically constant, like conventional GNNs. I investigate two equivariant geometric GNNs [Jing et al., 2021, Satorras et al., 2021], and an equivariant transformer [Liao and Smidt, 2023]. I plot the output distribution of random models on Erdős-Rényi graphs of different sizes in fig. 3, and find that all models tend to constant functions for large graphs.

EGNN tends to a constant function marginally slower than the other architectures. I conject that this is as side-effect of the continual normalisation and residual connections within it. The rate at which

all models converge to a constant is approximately the same as was reported by [Adam-Day et al., 2024] in their experiments on conventional GNNs. This suggests that the geometric problem setting does not meaningfully change the rate of convergence to a constant function.

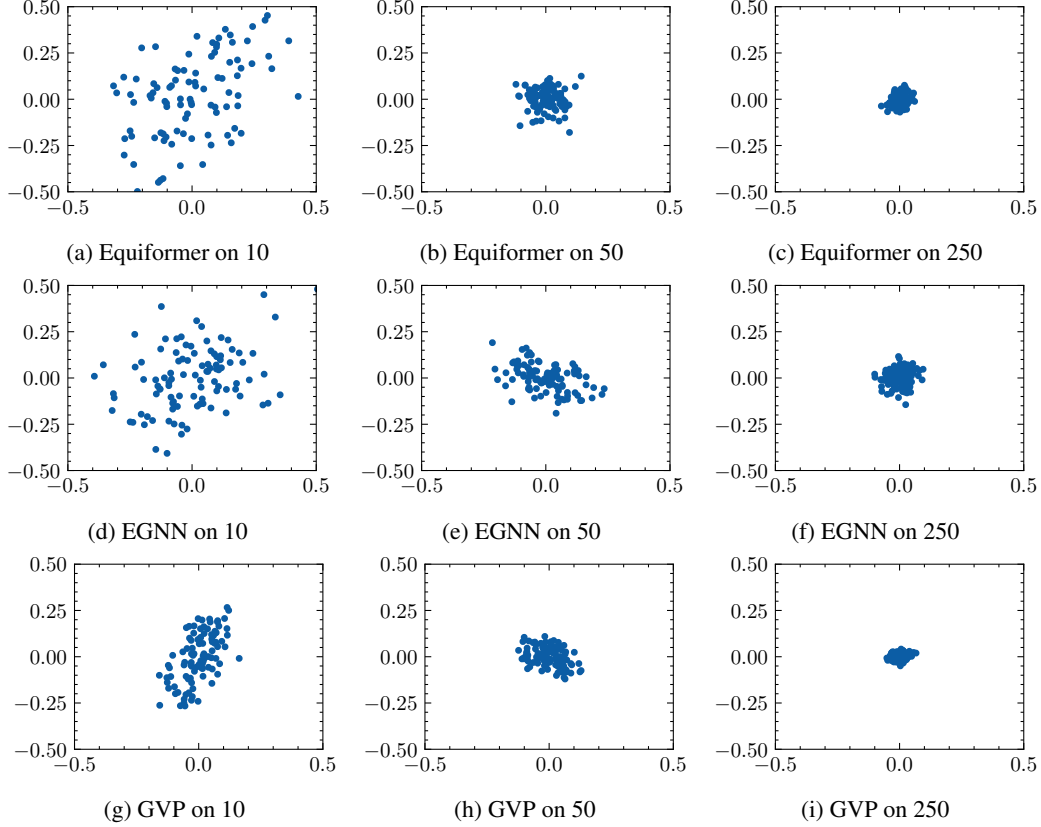


Figure 3: Output distribution of 4-layer models of different architectures on Erdős-Rényi graphs of different sizes with random features. Notice that all tested models collapse to constant functions as graph size increases. This reflects the theory developed by Adam-Day et al.. I find that EGNN [Satorras et al., 2021] collapses to a constant function the slowest, whilst GVP-GNN [Jing et al., 2021] collapses the fastest.

4.3 Modern architectures are no longer susceptible to oversmoothing

Oversmoothing has frequently been presented as the most fundamental limitation of GNNs. I find, however, that modern geometric GNN architectures have largely overcome it. I adapt the methodology of Rusch et al. to make it applicable to geometric GNNs, and find that none of the geometric GNN architectures I tested suffered noticeably from oversmoothing. In fig. 4, I plot the Dirichlet Energy of predictions of randomly initialised networks of different architectures on Erdős-Rényi graphs of different sizes as a function of model depth. I find that it is largely independent, which indicates that oversmoothing is not a significant problem on these modern architectures.

Modern equivariant models are not susceptible to oversmoothing I find that modern equivariant models implement many of the ‘solutions’ to oversmoothing as part of inductive biases. As a result, they are not prone to oversmoothing. Specifically, almost all modern models have residual connections, and frequently use layer normalisation, both of which are known to mitigate oversmoothing [Li et al., 2019, Rusch et al., 2023]. I conject that enforced equivariance with respect to geometric features helps force nodes to be differentiable, but am unable to prove this.

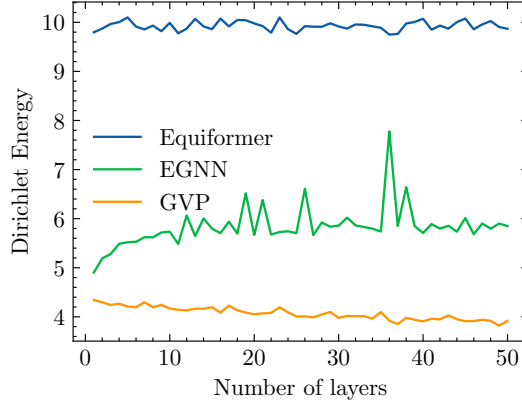


Figure 4: Graph comparing how prone to oversmoothing different architectures are. Dirichlet Energy is a measure of the variability of outputs, and we want it to remain constant with model depth. This indicates, similar to the results of Rusch et al., that modern message-passing architectures do not suffer from oversmoothing to the same extent that earlier architectures did.

4.4 Modern equivariant architectures are bounded by GWL, and invariant architectures are bounded by IGWL

Finally, I empirically test whether different geometric graph architectures can distinguish between the graphs presented in fig. 2 and present the results of my experiments in table 2. I find that, agreeing with the theory of [Joshi et al., 2023], the expressivity of equivariant networks is bounded by GWL, and the expressivity of invariant networks is bounded by IGWL.

Table 2: Results of my empirical tests on expressivity of geometric models. Models with a green background agree with the theory. Note that since Equiformer is an equivariant transformer, and is not based on the message-passing paradigm, prior expressivity research does not apply to it, and thus there are no formal bounds.

	Architectures				Equiformer
	GVP	Invariant GVP	EGNN	Invariant-EGNN	
$\mathcal{G}_1 \stackrel{?}{=} \mathcal{G}_2$	✓	✓	✓	✓	✓
$\mathcal{G}_1 \stackrel{?}{=} \mathcal{G}_3$	✓	✓	✓	✓	✓
$\mathcal{G}_2 \stackrel{?}{=} \mathcal{G}_3$	×	✓	×	✓	×

Of particular note, I find that Equiformer is unable to distinguish between the GWL indistinguishable graphs which I present it. This does not imply that the expressivity of Equiformer, or equivariant transformers, is upper-bounded by GWL, but rather that it is not significantly more expressive than GWL, and thus, the high performance of Equiformer is not as a result of expressivity.

5 How successful was the study?

This study was proposed with two objectives: understanding where recent improvements have come from; and helping to guide future model architects. The paper has made progress towards both of these goals, although both problems are large and still have many unanswered questions.

Where have recent improvements come from The two most surprising result of this paper are those of sections 4.1 and 4.4, that equivariant transformers are not significantly more expressive than geometric GNNs, yet are significantly better at modelling long-range dependencies. This implies that the performance gains of Equiformer, and other equivariant transformers may not be due to expressivity, but due to the powerful inductive bias of the transformer allowing better modelling of long-range dependencies. Note that these experiments do not rule out perceived performance gains

due to better machinery or hyperparameters, which is notoriously difficult, and often known to cause the misconception of progress [Liu et al., 2022, Tönshoff et al., 2024, Tori et al., 2025].

Guiding future model architects As observed in section 4.1, equivariant transformers are better at modelling long-range dependencies, and result in performance gains. This infers that future model architects may benefit from constructing models with more powerful inductive biases towards nodes which are further afield. Additionally, I found that all models tested appear to have overcome oversquashing through the use of normalisation and residual connections; this implies that there are not many possible performance gains by further addressing this problem.

The study, however, limits how generalisable it is by choosing to investigate older architectures. When designing novel architectures, researchers often generalise or modify existing architectures, such as Aykent and Xia did with the work of Cen et al.. By choosing to investigate older architectures, which have subsequently been built upon, it’s less helpful to model architects. The variety in model architecture which I investigated was limited, I did not investigate any scalarisation-based models, nor did I investigate any steerable models [Brandstetter et al., 2022], or any bleeding-edge equivariant models without using the Clebsch-Gordan tensor product. This prevents future model architects wishing to work with thees models from directly building on my work.

6 How could we strengthen these results?

Investigations on diverse datasets The primary limitation of the study is that experiments were performed on synthetic datasets. This was largely due to time and compute limitations, however, this makes it difficult to draw conclusive conclusions about how the problems identified would generalise to real-world tasks. Furthermore, I only investigated Erdős-Rényi graphs, and the results may differ on other types of graphs, *i.e.* Barabási-Albert graphs with strong networking features.

Prove our bounds are tight I demonstrated that equivariant transformers are not significantly more expressive than GWL in section 4.4, but did not upper bound their expressivity. To conclusively say that performance gains of Equivariant transformers are not due to higher expressivity, I would have to upper-bound their expressivity. Proving a concrete upper bound would be a strong result, but would require constructing and justifying a mathematical formalism for an generalised abstract equivariant transformer, before reasoning about its behaviour.

Transformer specific problems One part of this study was to investigate what the fundamental limitations of equivariant transformers are. I focused on investigating whether well-known problems in the graph domain still applied to them. I did not, however, investigate how known problems with the transformer architecture affected equivariant transformers specifically. The results would have been stronger if I’d identified novel problems and limitations, *i.e.* whether attention rank collapse affects graphs [Barbero et al., 2024], or how the limits of the computational model of transformers limit the expressivity of equivariant transformers on graphs [Weiss et al., 2021].

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